

A fellow team member Joseph is a weaker-skilled and less experienced software developer. They have authored the following python functions. Document them with comments throughout. Do not modify any of the following code beyond inserting comments.

```
def is_even(n):
    result = False
    for i in range(n):
        if result == False:
            result = True
        else:
            result = False
    return result

def absolute_value(x):
    values = []
    for i in range(-1000, 1001):
        if i == x:
            if x < 0:
                values.append(-x)
            else:
                values.append(x)
    if values:
        return values[0]
    return None

def count_vowels(s):
    vowels = "aeiouAEIOU"
    count = 0
    not_vowels = 0
    for ch in s:
        if ch in vowels:
            count += 1
        else:
            not_vowels += 1
    _ = not_vowels
    return count
```

```
def reverse_string(s):  
    out = ""  
    length = len(s)  
    for i in range(length):  
        out = out + s[length - 1 - i]  
    return out  
  
def get_max(lst):  
    sorted_lst = sorted(lst)  
    return sorted_lst[len(sorted_lst) - 1]
```

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 Run

```
def is_even(n):  
    # Start with False and toggle the value once for each number  
    # from 0 up to (but not including) n.  
    result = False  
  
    # Iterate n times.  
    for i in range(n):  
        # If the current result is False, switch it to True.  
        if result == False:  
            result = True  
        else:  
            # Otherwise switch it back to False.  
            result = False  
  
    # After toggling n times, return the final value.  
    # Even values of n will result in False, odd values in True.  
    return result  
  
def absolute_value(x):  
    # Create a List to hold the result if x is found.  
    values = []  
  
    # Search through all integers from -1000 to 1000.  
    for i in range(-1000, 1001):  
        # When the loop value matches x, determine its absolute value.  
        if i == x:  
            if x < 0:  
                # For negative numbers, negate x to make it positive.  
                values.append(-x)  
            else:
```

```
        # For zero or positive numbers, keep the value as-is.
        values.append(x)

# If a value was added to the list, return the first one.
    if values:
        return values[0]

# If x was outside the searched range, return None.
    return None


def count_vowels(s):
    # Define the characters that should be counted as vowels.
    vowels = "aeiouAEIOU"

    # Track the number of vowels encountered.
    count = 0

    # Track the number of non-vowel characters encountered.
    not_vowels = 0

    # Examine each character in the input string.
    for ch in s:
        if ch in vowels:
            # Increment the vowel counter when a vowel is found.
            count += 1
        else:
            # Increment the non-vowel counter otherwise.
            not_vowels += 1

    # Assign the non-vowel count to a throwaway variable.
    # This value is not used further.
    _ = not_vowels

    # Return the total number of vowels found.
    return count


def reverse_string(s):
    # Build the reversed string one character at a time.
    out = ""

    # Store the length of the input string.
    length = len(s)

    # Iterate once for each character position.
    for i in range(length):
        # Append characters from the end of the string toward the start.
        out = out + s[length - 1 - i]

    # Return the completed reversed string.
    return out
```

```
def get_max(lst):  
    # Create a sorted copy of the input list.  
    sorted_lst = sorted(lst)  
  
    # Return the final element, which will be the largest value  
    # after sorting in ascending order.  
    return sorted_lst[len(sorted_lst) - 1]
```

